Course Registration System

FIRST PROJECT

Lesson: Systems Analysis and Design

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Part 1 – Requirements Description with Use Cases

Use Case: Course Registration
Primary Actor: Student

Main Scenario

1. Student accesses the educational system and initiates the course registration process.
2. System retrieves and displays the registration calendar and the list of available course offerings for the current semester.
3. Student selects a specific course offering to add to their schedule.
4. System verifies that the requested course offering has available capacity.
5. System checks that the student has passed all prerequisite courses required for the selected course.
6. System checks that the student meets corequisite requirements for the selected course.
7. System checks that the selected course offering and its final exam do not have time conflicts with the student's existing schedule.
8. System verifies that the student has not previously passed this course.
9. System checks that adding this course does not cause the student to violate minimum or maximum credit limits based on their GPA.
10. System adds the course offering to the student's schedule and updates the credit total.
11. Student confirms and finalizes the semester's course registration.
12. System saves the registration details and displays the finalized schedule to the student.

Alternative Scenarios

In step 4 (course capacity full):
1. System displays an error message stating that the course capacity has been reached.
2. At the end, return to step 2 of the main scenario.

In step 5 (missing prerequisites):
1. System displays an error message identifying the missing prerequisites.
2. At the end, return to step 2 of the main scenario.

In step 6 (corequisite conditions not met):
1. System displays an error message stating that the required corequisites are not included.

2. At the end, return to step 2 of the main scenario.

In step 7 (time conflict with course offering or exam):

1. System displays an error message specifying the exact schedule conflict.
2. At the end, return to step 2 of the main scenario.

In step 8 (course already passed in previous semester):

1. System displays an error message indicating the course cannot be retaken because it has already been passed.
2. At the end, return to step 2 of the main scenario.

In step 9 (credit limit violation based on GPA):

1. System displays an error message specifying the credit limit violation.
2. At the end, return to step 2 of the main scenario.

Course Registration Domain Model